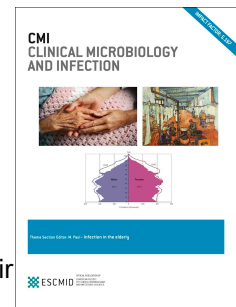


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Title:

Pandrug-resistant *Pseudomonas* spp. expresses New Delhi Metallo- β -lactamase-1 and consumes ampicillin as sole carbon source

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Abstract

Objectives

This study aims to investigate ampicillin catabolism in a pandrug-resistant strain, *Pseudomonas* sp. MR 02 of *P. putida* lineage.

Methods

The characterization of carbapenem-resistance was done following the standard protocol. The broth macrodilution method was used to determine the MIC values of antimicrobial agents both in the presence and in the absence of phenylalanine- β -naphthylamide (PA β N). High MIC value (>10000 mg/L) of ampicillin led to speculation that it may serve as a growth substrate, and thus minimal medium was used to evaluate ampicillin as a nutrient. The growth of MR 02 was measured in minimal medium in the presence or absence of 0.4 mM EDTA, supplemented with ampicillin as sole carbon, nitrogen, and energy source. RNA-seq was used to generate expression profiles of genes in ampicillin or glucose-grown cells. The *bla*_{NDM-1} gene of MR 02 was cloned in the pHSG398 vector and expressed in *E. coli* DH5 α .

Results

Phenotypic analysis along with genome sequence data identifies *Pseudomonas* sp. MR 02 as a pandrug-resistant strain. Transcriptome data has revealed that *bla*_{NDM-1} was among the top 50 differentially expressed genes in ampicillin grown cells compared to the glucose grown cells in the minimal medium. Heterologous expression of *bla*_{NDM-1} gene in *E. coli* DH5 α enabled its growth and subsistence on ampicillin as the sole source of carbon and energy.

Conclusions

The ability of a pandrug-resistant *Pseudomonas* sp. MR 02 to consume ampicillin for growth has a huge implication in the bioremediation of β -lactam residues in the environment.

Keywords: *Pseudomonas* spp., pandrug-resistant bacteria, antibiotic resistance, NDM-1 beta-lactamase, ampicillin

1. Introduction

Antibiotic resistance is one of the major risks to global health. *Pseudomonas* spp, belonging to gram-negative bacteria, known for their ubiquitous presence in soil and water, is also recognized to be associated with nosocomial infections by their multiple-antibiotic-resistant members [1]. Hence, there is a rising concern in exploring the occurrence of antibiotic resistance in environmental *Pseudomonas* spp. The aquatic environment is seen as the hot spot for the acquisition and spread of antibiotic resistance. The presence of extended-spectrum β -lactamases (ESBLs) and metallo- β -lactamases (MBLs) in river-water-isolates has important implications for humans [2]. *P. aeruginosa* is a well-known multidrug-resistant (MDR) pathogen [3,4], and also that environmental *Pseudomonas* such as *P. putida* and others are emerging with MDR profiles [5].

In the present study, an NDM-1 bearing river water isolate, *Pseudomonas* sp. MR 02, has shown a very high MIC value ($>10,000$ mg/L) of ampicillin. Earlier studies have found that hydrolyzed ampicillin binds to the zinc ions and various critical residues present in the active site of NDM-1 through a number of coordination bonds and hydrogen bonds. Also, NDM-1 has a high catalytic efficiency ($K_{cat}/K_m = 3.84 \mu M^{-1} S^{-1}$) of ampicillin hydrolysis [6, 7]. However, the mechanism of catalysis of ampicillin inactivation by NDM-1 remained obscure until combinational studies have proposed that active-site conformational fluctuation supports the enzymatic activity of NDM-1 which may lead additional studies about the catalytic mechanism [8]. As there was no certainty, but speculation (from extremely high MIC for ampicillin) that ampicillin may serve as any of the growth substrates, the minimal medium was used to evaluate ampicillin as a nutrient for MR 02. Subsequently, the growth curve of MR 02 in minimal medium devoid of glucose, but supplemented with ampicillin confirmed the utilization of ampicillin as the sole source of carbon and energy. In β -lactam antibiotic catabolism, amidases link resistance enzymes to central metabolism via phenylacetic acid catabolic pathway [9].

We hypothesized that possession and expression of NDM-1, in *Pseudomonas* sp. MR 02, would be the prime reason for its ability to tolerate high concentrations of ampicillin. Also surmised that to catabolize ampicillin, innate support of active amidases, and phenylacetic acid (*paa*) pathway genes should operate in the bacterium.

In this study, the molecular basis of ampicillin catabolism in an NDM-1 producing, pandrug-resistant river water isolate, *Pseudomonas* sp. MR 02 of *P. putida* lineage, has been examined in detail. The data generated from whole-genome-transcriptomics has provided evidence to support our hypothesis.

2. Materials and methods

Pseudomonas sp. MR 02 was isolated from a single sampling station on River Mahananda underneath the Mahananda bridge (26°43'11.52"N, 88°25'8.80"E) situated in the heart of Siliguri city, West Bengal, India.

Antibiotic susceptibility testing was performed by Kirby-Bauer disk diffusion method following CLSI guidelines and breakpoints [10]. Minimum inhibitory concentrations (MICs) of antimicrobial agents under different antimicrobial categories (including the ones as per international expert proposal for interim standard definitions for acquired resistance in case of *Pseudomonas aeruginosa*) were determined by a standard broth macrodilution method, in Muller-Hinton broth, following BSAC and EUCAST guidelines [11, 12, 13]. The influence of efflux pump on MIC values was determined by the broth macrodilution method in the absence or presence of a broad-spectrum inhibitor of RND pumps, namely Phe-Arg- β -naphthylamide (PA β N) at a concentration of 30 mg/L. The *bla*_{NDM-1} activity in MR 02 was determined by the broth macrodilution method in the presence of EDTA (0.4mM).

For whole genome sequencing, paired-end libraries of total DNA (isolated by phenol:chloroform isolation method) were generated by using Illumina Traseq Nano DNA library preparation kit and then sequenced by Next Seq 500 system. The genome sequence of MR 02 was analyzed for antibiotic-resistance genes (ARGs) using CARD database [14], followed by validation with BLAST analysis against information in the COG, NCBI, CDD and Swiss-Prot database. The MR 02 genome was compared with four other closely related *Pseudomonas* genomes by constructing the blast atlas using the BLAST Ring Image Generator (BRIG).

Growth of MR 02 was studied in minimal media containing ampicillin (6000 mg/L) as the sole source of carbon or both carbon and nitrogen. Whole transcriptome sequencing using Illumina Nextseq500 platform was done to compare the expression profile of genes in ampicillin-grown (test/ treated) with glucose-grown (control) MR 02 cells. The *lac* promoter-enabled *bla*_{NDM-1} (of MR 02) expression in *E. coli* DH5 α , using pHSG398 vector, was done to demonstrate growth

and subsistence in minimal media containing ampicillin (100 mg/L) as the sole carbon and energy source. [Details of the materials and methods are provided in the supplementary file.]

Funding

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3. Results

The MIC and disc diffusion test for strain MR 02 demonstrated resistance to 25 different antimicrobial agents belonging to 13 antimicrobial categories (Table S1 and S2). In terms of the criteria for defining MDR, XDR, and PDR [as per international expert proposal for interim standard definitions for acquired resistance in case of *Pseudomonas aeruginosa* (13)], it was found that *Pseudomonas* sp. MR02 was resistant to all agents in all antimicrobial categories (Aminoglycosides, Antipseudomonal carbapenems, Antipseudomonal cephalosporins, Antipseudomonal fluoroquinolones, Antipseudomonal penicillins + β -lactamase inhibitors, Monobactams, Phosphonic acids, and Polymyxins). Therefore, MR 02 is a Pandrug-resistant (PDR) strain.

In the presence of the efflux-inhibitor, PA β N, fold reduction in the MICs of chloramphenicol, tetracycline, ciprofloxacin, and levofloxacin were 50, 50, 75, and 6 respectively; thus demonstrating temporary reversion of the resistance to susceptible phenotype in the presence of PA β N. The MICs of imipenem, cefepime, and ceftazidime, in presence of PA β N, were reduced to a lesser degree (1.33 fold); and 1.63 fold reduction in the MIC of meropenem. (Fig. S1a). It was observed that in the presence of metal chelator, EDTA, the fold reduction in MICs of imipenem, meropenem, cefepime, and ceftazidime were 20.0, 5.26, 2.27, and 2.63 respectively (Fig. S1b).

The 16S rRNA gene sequence of *Pseudomonas* sp. MR 02 (Ac. No. MN252106) produced maximum identity (99.87%) with *P. monteilii* and *P. plecoglossicida*; 99.8% with *P. putida*; and 96.2% with *P. aeruginosa* in BlastN analysis. The overall similarity and the similarity of shared elements between two genomes were determined from the values obtained from *in silico* DNA: DNA hybridization (isDDH) and Average Nucleotide Identity (ANI) respectively. The isDDH and ANI values obtained for MR 02 with type strains, *P. monteilii* NBRC 103158^T or *P.*

plecoglossicida DSM 15088^T or *P. putida* NBRC 14164^T or *P. aeruginosa* DSM 50071^T were 40.5 & 89.9% or 33.1 & 86.9% or 40.4 & 90.% or 28.7 & 84.5%, respectively. Hence, the distinctiveness of MR 02 from the other species was established (because an *is*DDH of $\geq 70\%$ at the upper 95% confidence interval or an ANI value of $\geq 96\%$ denote genomes belonging to the same species) [15]. A visual analysis system was used for gene characterization, functional assignment and function-based browsing of the five genomes, MR 02, *P. monteilii* NBRC 103158, *P. plecoglossicida* DSM 15088, *P. putida* NBRC 14164, and *P. aeruginosa* PA01 RefSeq genome (Ac. No. PRJNA57495). Non-homologous genomic regions present in the genomes of the four different species of *Pseudomonas*, compared with MR 02, have been presented where a sub-set of it was related to antibiotic-resistance genes. Out of the predicted antibiotic resistance genes, 08 genes, extended-spectrum beta-lactamase (*bla*_{PME-1}), beta-lactamase (*bla*_{PDC}), tetracycline efflux MFS transporter Tet(G), TetR family transcriptional regulator, aminoglycoside 3-N-acetyltransferase [*aac*(3)-iiia], sulfonamide-resistant dihydropteroate synthase (*Sul1*), subclass B1 metallo-beta-lactamase (*bla*_{NDM-1}), and aminoglycoside nucleotidytransferase [*ant*(4)-IIb] were found unique to MR 02 (Fig. 1). Genome analyses identified a total of 98 genes associated with antibiotic resistance (Table S3). These genes are divided into two major groups coding for: (i) the enzymes that conferred antibiotic resistance (Table S4a) and (ii) the efflux pumps reported conferring antibiotic resistance (Table S4b). 64 genes have been assigned to 21 efflux pump systems belonging to five different transporter families. Three different genes coding for β -lactamases in the MR 02 genome, belonging to class A (PME-1), class B (NDM-1), and class C (AMPC) β -lactamase family.

The strain MR 02 has shown its subsistence in minimal media supplemented with 6000 mg/L ampicillin as the sole source of carbon and nitrogen at 30⁰C (Fig. 2a); while it grew well in 6000 mg/L ampicillin as the sole source of carbon, demonstrating exponential growth from 120th h to 168th h of incubation, followed by the stationary phase (Fig. 2b). Under similar culture conditions, MR 02 cells failed to subsist and grow in ampicillin as the sole carbon source in the presence of 0.4 mM EDTA (Fig. 2c); but its growth in minimal medium or rich medium, LB, remained almost unaffected in the presence of 0.4 mM EDTA (Fig. 2d, Fig. S2). Genomic analyses confirm the existence of the genes coding for enzymes and pathways utilized during β -lactam catabolism as per earlier report [9]. Three chromosomal β -lactamase genes, belonging to

the class A, subclass B1 and class C β -lactamase families and genes for complete phenylacetic acid catabolic pathway were annotated. Distribution of antibiotic-resistance genes (known to confer resistance to 14 different antibiotics) in the MR 02 genome is shown in Fig. S3. Another figure has been constructed with the annotated genes and operons for catabolism of β -lactam antibiotic, ampicillin (Fig. S4). RNAseq data of MR 02 cells (grown in minimal media containing ampicillin or glucose as carbon source) were used to analyze the expression of genes responsible for the catabolism of ampicillin. The expression of the *bla*_{NDM-1} gene was significantly upregulated ($\log_2$ fold change of 2.00088), representing one of the top 50 differentially expressed genes (Fig. S5)] while the expression of the other two β -lactamases, *bla*_{PME-1} and *bla*_{PDC}, in ampicillin-grown cells were similar to that of glucose-grown cells. Two genes, coding for penicillin G amidase and penicillin acylase family protein, and all *paa* genes of the phenylacetic acid catabolism were expressed in both conditions. On examining the expression profile of the *paa* genes, it was revealed that $\log_2$ fold change in the case of *paaB* and *paaG* genes were 0.93659 and 0.528505 respectively (higher expression in ampicillin-grown cells compared to glucose-grown cells, but not 'upregulated') (Table S5a & b).

As both physiological and transcriptome data have indicated the pivotal role of NDM-1 in supporting the growth of MR 02 using ampicillin as sole carbon and energy source, it was hypothesized that this phenomenon should work in *E. coli* having similar metabolic pathways. Hence the NDM-1 gene of MR 02 was cloned and expressed in *E. coli* DH5 α . The recombinant strain of *E. coli* DH5 α was able to grow marginally in ampicillin (100 mg/L) as the sole carbon source (Fig. S6), thereby supporting the associated role of NDM-1 in empowering bacteria to consume β -lactams for subsistence.

4. Discussion

This study describes ampicillin catabolism in a pandrug resistant *Pseudomonas* sp. MR 02. While determining the MICs of different antibiotics, including different types of β -lactams, very high MIC values (mg/L) for cephalosporin (cefepime and ceftazidime, 1600), carbapenem (imipenem, 160; meropenem, 650), and ampicillin (>10000) were obtained. The resistance phenotype against the β -lactam antibiotics was corroborated by the genomic presence of three different β -lactamases, PME-1, NDM-1, and AMPC. Indeed, MR 02 has surpassed *P. aeruginosa* PA01 in acquiring antimicrobial-resistance genes with 08 more ARGs. To explain the genetic basis of MR 02's efflux-dependent resistance against chloramphenicol, tetracycline,

ciprofloxacin, and levofloxacin 8, 11, 5, and 5 efflux-system genes/gene-clusters were annotated from its whole-genome-sequence (Fig. S3). Furthermore, 14 antibiotic-related genomic islands were identified using island viewer (Fig. S7).

We hypothesized that the possession of NDM-1 could be the primary reason for yielding high MIC value of ampicillin, and consumption of ampicillin as a sole source of carbon and energy. It was reported earlier that NDM-1 displays high catalytic efficiency of ampicillin hydrolysis ($K_{cat}/K_m = 3.84 \mu M^{-1} S^{-1}$), and unlike other β -lactamases, rather localizes itself to the inner leaflet of the outer membrane and secreted via outer membrane vesicles [7, 16, 17]. Therefore, these properties of NDM-1 may have given a selective advantage to the bacterium to grow using ampicillin as the sole source of carbon and energy. Earlier studies have shown that soil bacteria can survive in minimal media containing antibiotic, either natural or artificial, as the carbon source [18]. Many of these antibiotics are produced by soil bacteria. The concentration of such antibiotics could be in milligram quantity per gram of soil [19]. Hence, it is predictable that certain groups of bacteria might have developed a mechanism for survival by consuming antibiotics as the carbon source. A report on β -lactam catabolism (penicillin) [9] has additionally supported the hypothesis. In fact, MR 02 has shown an extraordinary ability to grow in the high concentration of ampicillin. The role of NDM-1 in supporting the growth of MR 02 on ampicillin, as sole carbon, and energy source, was established because the growth in the same medium was inhibited in the presence of a metal scavenger, EDTA. NDM-1 is known to be more susceptible to EDTA, than other metallo- β -lactamases; and 50% inhibition of NDM-1 activity was reported in the presence of 412 nM EDTA [20]. It was observed that EDTA had no significant effect on the growth of MR 02 in unamended minimal medium or Luria broth. Despite having a 6-aminopenicillanic acid centre containing nitrogen in most β -lactam antibiotics, ampicillin owns a side chain having another amino group. It had been observed that ampicillin was also used by MR 02 as the only source of both nitrogen and carbon source. It's very likely that in a new growth environment, where no readily metabolizable carbon source such as sugar is available, MR 02 faces a survival challenge. Therefore, the bacterium might be requiring a protracted lag period to adapt and synthesize enzymes to utilize the alternate carbon source, ampicillin, with the assistance of NDM-1. Moreover, in support of the hypothesis, we have presented the results regarding log c.f.u/mL vs time (0h to 48h) to show a rise in the viable

cell number of *E. coli* DH5 α :pHSG398:*bla*_{NDM-1} using ampicillin (100 mg/L) as the only source of carbon and energy.

Based on the comparative transcriptomics, it was demonstrated that besides the expression of genes coding for β -lactamases, amidases, and *paa* pathway, there has been a change in the expression of metabolic enzymes to meet growth requirements in minimal medium containing ampicillin as sole carbon and energy source. MR 02 cells may have used the gluconeogenesis pathway to synthesize glucose-6-phosphate to create various cell macromolecules (as glucose wasn't available in the medium). The expression of *pck*, *gapN*, *pgm*, *adhC*, *gpml*, and *tpiA* were higher along with significant upregulation of *aldB* (coding for aldehyde dehydrogenase) and *exaA* (coding for PQQ dependent dehydrogenase) genes of the gluconeogenesis pathway (Table S5a). Moreover, three genes of the TCA cycle coding for NADP-dependent isocitrate dehydrogenase, 2-oxoglutarate dehydrogenase E1 component (*sucA*), succinate dehydrogenase flavoprotein subunit (*sdhA*, *frdA*); and one gene coding for D-glycerate dehydrogenase (phosphor-gluconate 2-dehydrogenase) of pentose phosphate pathway were found to be significantly upregulated, suggestive of regulation of genes related to growth on ampicillin as sole carbon and energy source. Two genes coding for NAD(P)H dehydrogenase (quinone) of ubiquinone and terpenoid-quinone biosynthesis pathway was significantly downregulated (Table S5b).

The results presented up to now in this research have supported the hypothesis about the function of NDM-1 in the catabolism of ampicillin. However, there are limits of this study because no knockouts of these genes have been made to translate genome and transcriptome data in the context of ampicillin catabolism. Moreover, data from proteomics and metabolomics were not generated as a means to integrate them with genome and transcriptome data. Further research in this direction will elucidate novel bacterial strategies in repurposing metabolic pathways to derive nourishment from inactivated antibiotics.

Nucleotide sequence accession numbers

The GenBank accession number for the 16S rRNA gene sequence of MR 02 is MF401548.1. The whole-genome shotgun project has been deposited at DDBJ/ENA/GenBank under the accession number PESJ000000000. The transcriptome data have been deposited in NCBI under BioProject number PRJNA612657 with BioSample numbers SAMN14379771 and SAMN14379772.

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Ethical approval

Not required

Supplementary material

Transparency declaration

The authors declare no conflicts of interest.

Authors' contribution

R.C. developed the study, designed the experiment, analyzed the transcriptome data, and wrote the manuscript. V.K.R. anchored the main study along with molecular biology work, performed the laboratory experiments with K.G. S.M. interpreted the results and took part in paper writing. S.T. performed all bioinformatic analyses. All authors gave feedback and participated in the manuscript editing.

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Figure Legends

Figure 1: Circular representation of the *Pseudomonas* sp. MR 02 draft genome in comparison with four reference genomes of *Pseudomonas* genus. Ring from inside to outside (1) GC contents (black) (2) GC skew (purple and green) (3) Coding sequences of *Pseudomonas* sp. MR 02 draft genome (dark blue) (4) BLAST comparison with *P. plecoglossicida* DSM 15088 (red) (5) BLAST comparison with *P. monteilii* NBRC 103158 (green) (6) BLAST comparison with *P. putida* NBRC14164 (violet) (7) BLAST comparison with *P. aeruginosa* PA01 (lime green) (8) Position of some important antibiotic resistance genes (red). Figure generated by BLAST Ring Image Generator v 0.95.

Figure 2 (a-d): Consumption of ampicillin as growth substrate by *Pseudomonas* sp. MR 02

- (a) Growth of MR 02 in ampicillin (6000 mg/L) as sole carbon, nitrogen, and energy source
- (b) Growth of MR 02 in ampicillin (6000 mg/L) as sole carbon and energy source
- (c) Growth of MR 02 in ampicillin (6000 mg/L) as sole carbon and energy source in the presence of metal-chelator, EDTA (40 mM)
- (d) Growth of *Pseudomonas* sp. MR 02 in minimal media [containing glucose (6000 mg/L) as sole carbon and energy source and NH_4Cl (400 mg/L) as sole nitrogen source] in the presence or absence of EDTA (40 mM)

Figure 1:

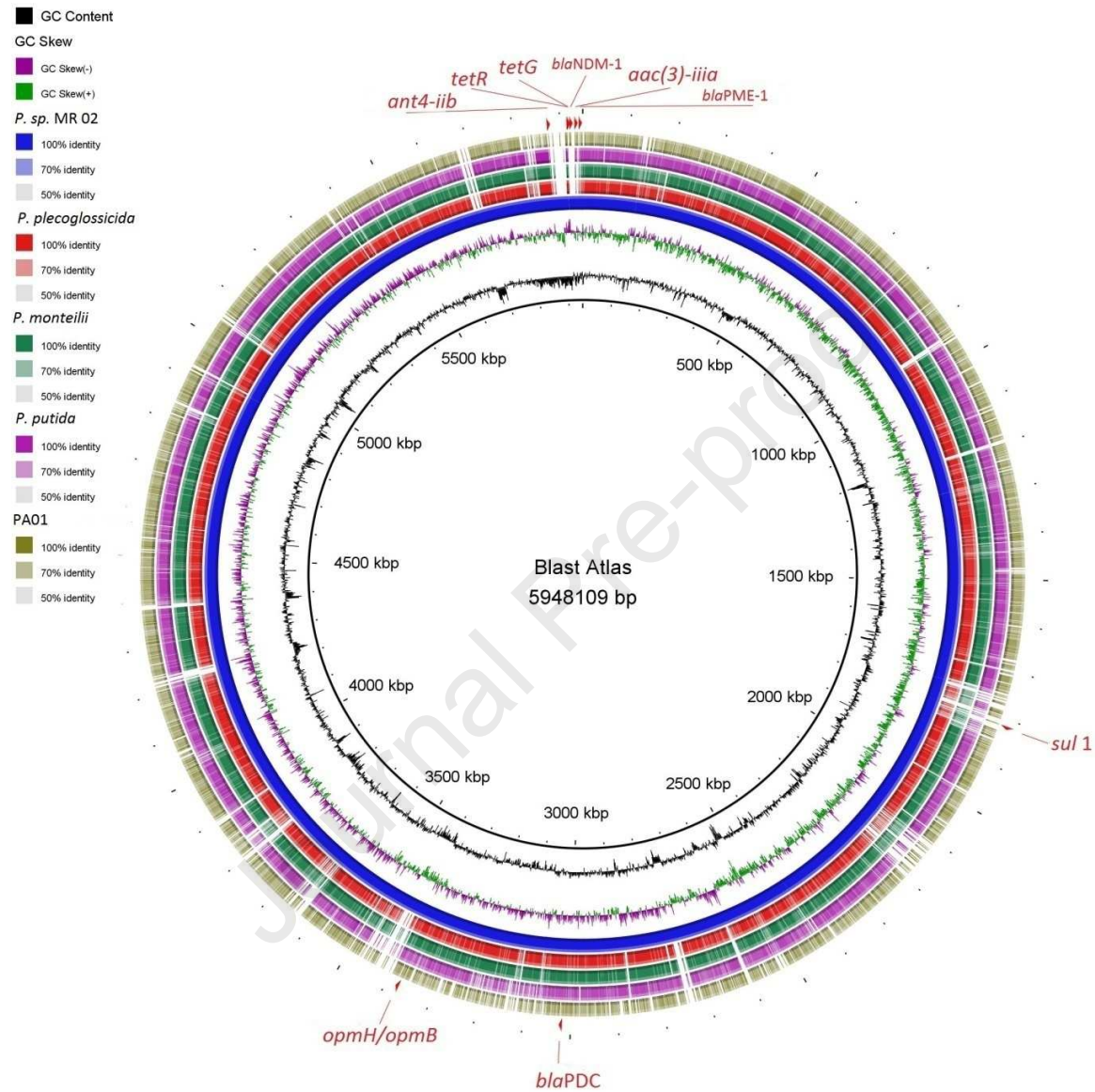


Figure 2:

